A rep of a MLG G is a group homomorphism

$\rho : G \to \underline{\mathrm{Aut}(V)}$ where V is a complex Hilbert space

$\to$ invertible linear maps $V \to V$

ρ maps $g \in G$ to the action of g on V

$\rho(e) = \mathrm{id}$

$\rho(g_1 g_2) = \rho(g_1) \circ \rho(g_2)$

(also want it to be <u>smooth</u>) $\to$ how?

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in terms of group actions: if ρ is a rep. then

$$\boxed{\begin{array}{ccc} G \times V & \to & V \\ g \in v & \mapsto & \rho(g)(v) \end{array}}$$ is a group action

$\to$ ask that this is smooth

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ex trivial representation : V is one-dimensional $\rho(g) = \mathrm{id} \quad \forall g \in G$

ex $G = U(1)$ $\quad V = \mathbb{C}$

$\forall n \in \mathbb{Z}$ define $\rho_n : g \in U(1) \mapsto \rho_n(g) \in \mathrm{Aut}(\mathbb{C})$

$$\rho_n(g)(z) = g^n z$$

- action is linear
- $\rho_n(1)(z) = z \quad \forall z \Rightarrow \rho_n(1) = \mathrm{id}$
- $\rho_n(g_1 g_2)(z) = (g_1 g_2)^n z = g_1^n g_2^n z = g_1^n (g_2^n z) = g_1^n \rho_n(g_2)(z) = \rho_n(g_1)(\rho_n(g_2)(z))$

check $\rho_n(e^{\theta + 2\pi k}) = \rho_n(e^{\theta})$ ok because $n \in \mathbb{Z}$

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the same works if $V = \mathbb{C}^N$ $\quad \rho_n(g)(z) = g^n z \quad (z \in \mathbb{C}^N)$

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also bounded

- $\rho: G \to \mathrm{Aut}(V)$ is unitary if $\rho(g)$ is a unitary map $\forall g$

$\to$ means that $\langle \rho(g)v, \rho(g)w \rangle = \langle v, w \rangle \quad \forall g \in G, \forall v, w \in V$

alternatively $\rho(g)^\dagger = \rho(g)^{-1} = \rho(g^{-1}) \quad \| \quad \langle \rho(g)^\dagger v, w \rangle = \langle v, \rho(g) w \rangle$

$L: V \to V$ bounded if $\exists M > 0$ s.t. $\|L(v)\| \leq M \|v\| \quad \forall v \in V$

- <u>irreducibility</u> $\rho: G \to \mathrm{Aut}(V)$ is irreducible if the only subspaces of V that are closed under the action of $\rho(g) \forall g$ are $\{0\}$ and V itself

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if $W \subseteq V$ and $\rho(g) w \in W \quad \forall w \in W, g \in G$

then $\tilde{\rho}: g \in G \to \rho(g)|_W \in \mathrm{Aut}(W)$ subrepresentation

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<u>Theorem</u> if G is compact, then all unitary irreducible representations are finite-dimensional.